

Series and parallel: where the gauge insertion law stops closing

Result, 2026-09-26. Halve one lattice direction of a heat-kernel lattice gauge theory in D dimensions. The step splits exactly into two kinds of move. Faces containing the refined direction are cut in two; these are *series* moves, and the heat-kernel convolution integrates them out exactly in every dimension. Faces transverse to the refined direction acquire a new parallel copy at doubled heat time; these are *parallel* moves, and integrating out the new copy leaves the factor Ψ of Proposition 1: the partition function of a two-dimensional gauge theory on the inserted mid-plane, whose edges carry Brownian-bridge laws fixed by the coarse field instead of Haar measure. The number of transverse planes is $\binom{D-1}{2}$: none in $D = 2$, which is why two-dimensional Yang–Mills closes; one in $D = 3$; three in $D = 4$. In the free (abelian Gaussian) theory Ψ is computed exactly (Proposition 2): the defect against the coarse action is a negative semidefinite quadratic form of relative size $a^2 K^2/16 + (Ga)^2/8$ for fields with transverse momenta at most K and layer gradient G . It is the field-theory counterpart of Newton’s $-F^2_{uvh}/(8M)$ in the refinement note, eq. (1), and it equals exactly the error of Migdal’s bond-moving approximation in the Gaussian theory. For a single cube with $G = U(1)$ the parallel move is exact in closed form (Proposition 3): a heat kernel at softened heat time on the interpolated flux, times a theta-function factor that is the large-field term with rate $e^{-2\pi(\pi-|\phi|)/t}$. The non-abelian estimate needed for one step in $1 + 2$ dimensions is stated in §4 as Hypothesis P(α), with the four terms it must control and their orders in $\lambda_3 a$.

This answers the stop rule of the refinement note, §8, in its second form: it gives the exact term that prevents closure, with an explicit free-field estimate, and states the interacting estimate that remains. The ingredients are established (heat-kernel convolution, Brownian bridges, Gaussian integration); the decomposition and the exact free-field defect are elementary, and no novelty is claimed for them beyond their use here as the organizing comparison. Migdal’s recursion (1975) is the approximate form of the same series/parallel split, as Kadanoff’s notes describe it (Kadanoff 1976, metadata).

1. Heat times on an anisotropic lattice

Take a hypercubic lattice in D dimensions with spacings $a_1, \dots, a_D$, compact gauge group G with the group metric of the refinement note §5 ($-\Delta_G \chi_R = C_2(R) \chi_R$, $C_2(\mathbf{3}) = 4/3$ for $SU(3)$), and heat kernel $k_t = e^{t\Delta_G/2}$, $k_t(U) = \sum_R d_R \chi_R(U) e^{-tC_2(R)/2}$. Use the dimensionally explicit continuum action of the refinement note §6,

$$\frac{S_E}{\hbar} = \frac{1}{4\lambda_D} \int F_{\mu\nu}^a F_{\mu\nu}^a d^D x, \quad [\lambda_D] = \text{length}^{D-4},$$

so that $\lambda_D = \hbar g_{\text{cl},D}^2$ contains $\hbar$. The anisotropic heat-kernel action assigns to a plaquette p in the plane $(\mu\nu)$ the weight $k_{t_{\mu\nu}}(U_p)$ with

$$t_{\mu\nu} = \lambda_D \frac{a_\mu a_\nu}{\prod_{\rho \neq \mu, \nu} a_\rho}. \quad (1)$$

Derivation of (1). For a smooth field, $U_p = \exp(a_\mu a_\nu F_{\mu\nu} + O(a^3))$ and, for small t away from the cut locus, $-\log k_t(e^X) = |X|^2/(2t) + \text{const} + O(t + |X|^2)$. Matching $a_\mu^2 a_\nu^2 |F_{\mu\nu}|^2/(2t_{\mu\nu})$ to the continuum density $(1/2\lambda_D) |F_{\mu\nu}|^2 \prod_\rho a_\rho$ (the factor $1/2$ collects $F_{\mu\nu}$ and $F_{\nu\mu}$) gives (1). $\square$

The heat time is dimensionless. In $D = 2$ it is λ_2 times the face area, as in the refinement note; in $D = 3$ the isotropic value is $t = \lambda_3 a$, the refinement parameter of that note’s table; in $D = 4$ it is $t = \lambda_4 = g_{\text{lat}}^2$, independent of a . Equation (1) is a choice of regularization, the heat-kernel counterpart of the anisotropic Wilson action; the theorems below concern it exactly.

2. Halving one direction: series moves and a parallel insertion

Halve direction 1: $a_1 \mapsto a_1/2$. By (1),

- a plaquette in a plane (1ν) has $t_{1\nu} \mapsto t_{1\nu}/2$, and each coarse (1ν) face is cut by a new edge into two fine faces;
- a plaquette in a plane (jk) with $j, k \neq 1$ has $t_{jk} \mapsto 2t_{jk}$; the old (jk) faces survive with doubled heat time, and a new (jk) face appears in each mid-plane $x_1 = (\ell + \frac{1}{2})a_1$, also at heat time $2t_{jk}$.

The new variables are the half-edges in direction 1 and the *mid-edges* lying in the mid-planes. Each mid-edge in direction $j \neq 1$ lies in exactly one coarse face of the plane $(1j)$, which it cuts in two, and it borders $2(D-2)$ mid-faces.

Proposition 1 (exact pushforward). Let μ' be the fine heat-kernel measure and p the projection multiplying half-edges. Then $p_*\mu'$ has density, with respect to the product Haar measure on coarse links,

$$\frac{1}{Z'} \prod_{f \ni 1} k_{t_f}(U_f) \prod_{f \perp 1} k_{2t_f}(U_f) \Psi(U), \quad (2)$$

where $f \ni 1$ ranges over coarse faces containing direction 1, $f \perp 1$ over the others, and

$$\Psi(U) = \int \prod_{g \text{ mid}} k_{2t_g}(m_{\partial g}) \prod_{e \text{ mid}} \beta_e(dm_e | U). \quad (3)$$

Here $m_{\partial g}$ is the ordered product of mid-edge variables around the mid-face g , and β_e is the midpoint law of the Brownian bridge on G of duration t_{1j} determined by the coarse face that e cuts: if the two halves of that face have holonomies $P_e m_e^{-1}$ and $m_e Q_e$, with $P_e Q_e = U_f$,

$$\beta_e(dm | U) = \frac{k_{t_{1j}/2}(P_e m_e^{-1}) k_{t_{1j}/2}(m Q_e)}{k_{t_{1j}}(U_f)} dm. \quad (4)$$

Proof. At each new vertex on a direction-1 edge, only the two half-edges and the incident mid-edges meet. Changing variables to (coarse product, upper half) and using Haar invariance, the gauge transformation at that vertex sets the upper half-edge to the identity without changing the integral; the lower half-edge becomes the coarse link. The half-faces then have the holonomies displayed, with P_e, Q_e built from coarse links only. Multiply and divide by $k_{t_{1j}}(U_f)$ for each cut face. By the convolution identity, eq. (9) of the refinement note, each β_e is a probability measure. Fubini gives (2)–(3). $\square$

The dimension count. Refining one direction produces parallel insertions in the $\binom{D-1}{2}$ planes transverse to it: zero for $D = 2$, one for $D = 3$, three for $D = 4$. With no transverse plane, $\Psi = 1$ and (2) is the exact consistency (7) of the refinement note, recovering its §5. An isotropic dyadic step is D successive directional halvings. By (1) the heat times change as follows:

D	one directional halving	full isotropic step	parallel planes per halving
2	$t \mapsto t/2$	$t \mapsto t/4$ (area)	0
3	$t_{1\nu}/2, 2t_{23}$	$t \mapsto t/2$	1
4	$t_{1\nu}/2, 2t_{jk}$	$t \mapsto t$	3

The last column locates the whole renormalization problem. The middle column is the classical scaling: superrenormalizable in $D = 3$, where the heat time halves per step, and marginal in $D = 4$, where it returns to itself and any running must come from Ψ .

The structure matches Newton's insertion. There the new time splits one cell in series and the elimination leaves the endpoint-independent scalar $-F^2 uvh/(8M)$; here the series moves close with no remainder (the heat kernel is already the exact propagator, the counterpart of U_h rather than Q_h), and the parallel insertion leaves Ψ . A Wilson action would also leave series remainders, as Q_h does.

3. The free field: the parallel defect in closed form

Take $G = \mathbb{R}^n$ with the Euclidean metric (the Gaussian limit of a compact group of dimension n ; $n = 8$ for $SU(3)$) and $D = 3$. Then $k_t(x) = (2\pi t)^{-n/2} e^{-|x|^2/(2t)}$, and β_e is Gaussian with mean $(P_e - Q_e)/2$ in additive notation and covariance $(t_{1j}/4) I_n$, independent of the coarse data. Put the lattice on a torus with $N_2 \times N_3$ transverse sites and N_1 layers. Write $\Phi_\ell(x)$ for the flux of the old (23) face at layer ℓ and transverse site x , and $\hat{\Phi}_\ell(k)$ for its unitary discrete Fourier transform in (x_2, x_3) , with $s_j = \sin(k_j a_j/2)$ and

$$v(k) = t_{12} s_3^2 + t_{13} s_2^2.$$

On a torus the transverse zero mode of a lattice curl vanishes, so $k \neq 0$ throughout. All statements concern gauge-invariant functions, which depend on the links only through the fluxes; the non-compact gauge orbits drop out with the normalization constants.

Proposition 2 (exact free-field parallel defect). In this Gaussian theory, the pushforward (2) is the coarse heat-kernel measure multiplied by $e^{-\mathcal{D}}$ (up to a constant), with

$$\mathcal{D} = - \sum_{\ell} \sum_{k \neq 0} \left[\frac{v(k)(|\hat{\Phi}_\ell|^2 + |\hat{\Phi}_{\ell+1}|^2)}{8t_{23}(2t_{23} + v(k))} + \frac{|\hat{\Phi}_\ell - \hat{\Phi}_{\ell+1}|^2}{8(2t_{23} + v(k))} \right] \leq 0. \quad (5)$$

Proof. (i) The mean mid-edge configuration interpolates linearly between layers ℓ and $\ell + 1$; the direction-1 links enter it as a lattice gradient, which the curl annihilates. Hence the mean mid-face flux is exactly $(\Phi_\ell + \Phi_{\ell+1})/2$. (ii) The fluctuation of the mid-face flux is the lattice curl of independent edge noises with variances $t_{12}/4$ (direction 2) and $t_{13}/4$ (direction 3). Its Fourier covariance is $(t_{12}/4)|1 - e^{ik_3 a_3}|^2 + (t_{13}/4)|1 - e^{ik_2 a_2}|^2 = v(k)$, diagonal in k . (iii) For Gaussian η of variance v , $E e^{-|\hat{\Phi} + \eta|^2/(2\sigma^2)} \propto e^{-|\hat{\Phi}|^2/(2(\sigma^2 + v))}$; with $\sigma^2 = 2t_{23}$, $-\log \Psi = \sum_{\ell, k} |\hat{\Phi}_\ell + \hat{\Phi}_{\ell+1}|^2 / (8(2t_{23} + v)) + \text{const.}$ (iv) The old (23) faces contribute $\sum |\hat{\Phi}_\ell|^2 / (4t_{23})$, the coarse target is $\sum |\hat{\Phi}_\ell|^2 / (2t_{23})$, and the cut faces return $k_{t_{1j}}(U_f)$ exactly. Subtract, and use $|A + B|^2 = 2|A|^2 + 2|B|^2 - |A - B|^2$. $\square$

At zero transverse momentum and constant flux in ℓ , the two old and new faces at heat time $2t_{23}$ combine in parallel to one face at t_{23} , as conductances do. The defect measures the two departures: transverse lattice momentum ($v > 0$) and variation between layers.

Corollary (size of one step). On an isotropic coarse lattice ($a_j = a$, $t_{\mu\nu} = t = \lambda_3 a$), suppose $\hat{\Phi}_\ell$ is supported on $|k| \leq K$ and $\sum_{\ell} \|\Phi_\ell - \Phi_{\ell+1}\|^2 \leq (Ga)^2 \sum_{\ell} \|\Phi_\ell\|^2$, with $\|\cdot\|$ the ℓ^2 norm over transverse sites. Then

$$|\mathcal{D}| \leq \left(\frac{a^2 K^2}{16} + \frac{(Ga)^2}{8} \right) \sum_{\ell} \frac{\|\Phi_\ell\|^2}{2t}. \quad (6)$$

Proof. Use $2t + v \geq 2t$ and $v \leq ta^2 |k|^2/4$ in (5). $\square$

For a smooth field, $\Phi = a^2 F_{23}$ and the right side of (6) is $a^2(K^2/16 + G^2/8)$ times the continuum magnetic action $(1/2\lambda_3) \int |F_{23}|^2 d^3x$: a dimension-six correction with coefficient a^2 . At fixed physical K and G the relative defects along $a_n = 2^{-n} a_0$ are summable, which is the free-field instance of the refinement note's criterion (12), in a norm on Gaussian actions rather than total variation. Two cautions keep this honest. After one step the effective action is (2) with $\mathcal{D}$, so the next step starts from a slightly different action; the free field is controlled exactly because every stage is Gaussian, and its blocked actions converge to a fixed point, the perfect action of Hasenfratz and Niedermayer (1994) (metadata). And (6) holds for smooth coarse data; typical lattice fields at fixed t are rough, so (6) states the irrelevance of the defect, while the measure-level estimate needs the whole quadratic form (5).

Migdal's bond moving, read exactly. Moving each mid-face onto the old face of layer ℓ multiplies two Gaussian weights at $2t_{23}$ into one at t_{23} , which is the coarse target. So in the Gaussian theory $\mathcal{D}$ is exactly the error of the bond-moving step, and (5) splits it into its noise part

(v) and its displacement part ($\Phi_\ell - \Phi_{\ell+1}$). Tomboulis's attempt to bound the exact decimation between Migdal–Kadanoff-type approximations (arXiv:0707.2179, abstract; not established as a theorem, see what-would-unblock) worked on the same parallel step.

The comparison with Newton. Both eliminations complete a square, and both lower the action. Newton's defect is a constant, removed by one cubic counterterm, after which the cell law closes exactly. The free-field defect is an irrelevant quadratic form; no finite set of nearest-neighbour counterterms removes it, and the closed object is the fixed point in the space of all quadratic actions. In the interacting theory Ψ is non-Gaussian.

4. The non-abelian parallel move in 1 + 2 dimensions

For compact G , (3) says: $\Psi(U)$ is the partition function of a two-dimensional heat-kernel lattice gauge theory on the mid-planes, at heat time $2t_{23}$, in which each edge is distributed by the bridge law $\beta_e(\cdot | U)$ instead of Haar measure. Two limits are exactly soluble. As $t_{1j} \rightarrow \infty$ the bridge laws tend to Haar measure, the mid-plane is the exactly soluble two-dimensional theory of the refinement note §5, and Ψ is a constant: the old transverse faces stay at heat time $2t_{23}$. As $t_{1j} \rightarrow 0$ the bridge laws concentrate at the geodesic interpolation and Ψ is the product of mid-face weights on the interpolated holonomies, a classical evaluation, like Newton's z_* . Between those limits, two-dimensional exact solvability fails, because it rests on Haar invariance of every edge.

Relative to the Gaussian computation of §3, four terms appear, with the orders that power counting assigns them at $t = \lambda_3 a \ll 1$ (labelled as power counting; proving them is the content of the hypothesis below):

- **N1, interpolation.** The mean mid-face holonomy differs from the abelian average by Baker–Campbell–Hausdorff commutators of the fluxes, of order $a^4[F, F]$ in $\log U_g$. In the action this is a dimension-six operator, $O(a^2)$ relative, like (6).
- **N2, bridge shape.** The bridge law depends on the coarse face holonomy through the group's curvature (the Jacobian of the exponential map in the small- t heat kernel). Relative size $O(t) = O(\lambda_3 a)$; gauge invariance allows it to shift the coupling and the vacuum energy only, at leading order.
- **N3, non-Gaussian fluctuation.** The mid-face holonomy is a non-commutative product of noisy edges; the connected corrections to $-\log \Psi$ are $O(t)$ relative and local.
- **N4, large fields.** Bridges that wind around the group (the image terms of the heat kernel near the cut locus) contribute $O(e^{-c/t})$ per face, non-perturbatively small at each step but to be controlled uniformly in the volume.

The compact abelian cube isolates N4 exactly. Take $G = U(1)$ with $k_t(\theta) = \sum_{n \in \mathbb{Z}} e^{-tn^2/2} e^{in\theta}$ (the Villain weight), and refine a single cube: one mid-face at heat time t_m , bounded by four mid-edges m_j with orientations $\sigma_j = \pm 1$, each cutting a side face of heat time t_j whose halves have angles $p_j - m_j$ and $m_j + q_j$. Put $\phi_j = p_j + q_j$ (the side-face flux, any real representative), $\bar{\Phi} = \sum_j \sigma_j (p_j - q_j)/2$ (the interpolated mid-face flux, with the same representatives), and

$$\rho_t(\phi) = \frac{\sum_{w \in \mathbb{Z}} (-1)^w e^{-(\phi - 2\pi w)^2/(2t)}}{\sum_{w \in \mathbb{Z}} e^{-(\phi - 2\pi w)^2/(2t)}} \in [-1, 1].$$

Proposition 3 (exact $U(1)$ cube). With $T = t_m + \frac{1}{4} \sum_j t_j$,

$$\Psi = \sum_{n \text{ even}} e^{-Tn^2/2} e^{in\bar{\Phi}} + \left(\prod_{j=1}^4 \rho_{t_j}(\phi_j) \right) \sum_{n \text{ odd}} e^{-Tn^2/2} e^{in\bar{\Phi}}. \quad (7)$$

If every $\rho_{t_j}(\phi_j) = 1$, then $\Psi = k_T(\bar{\Phi})$ exactly. For $|\phi_j| \leq \pi$ and $t_j \leq 1$, $0 \leq 1 - \rho_{t_j}(\phi_j) \leq 5e^{-2\pi(\pi - |\phi_j|)/t_j}$.

Proof. The bridge numerator for $E e^{inm}$ is $\sum_r e^{-t(r^2+(r-n)^2)/4} e^{irp} e^{i(r-n)q}$ by character orthogonality, and $r^2 + (r-n)^2 = 2(r-n/2)^2 + n^2/2$. For even n the shift $r \mapsto r + n/2$ reproduces the denominator $k_t(\phi)$, giving $E e^{inm} = e^{-tn^2/8} e^{in(p-q)/2}$. For odd n the shift is by a half-integer and leaves the ratio $\sum_s e^{-t(s-\frac{1}{2})^2/2} e^{i(s-\frac{1}{2})\phi} / \sum_s e^{-ts^2/2} e^{is\phi}$, which Poisson summation turns into $\rho_t(\phi)$. The four m_j are independent, so $E k_{t_m}(\sum_j \sigma_j m_j)$ is the sum over n of $e^{-tmn^2/2} \prod_j E e^{in\sigma_j m_j}$, which is (7). For the bound, write $G_w = e^{-(\phi-2\pi w)^2/(2t)}$; then $1 - \rho = 2 \sum_{w \text{ odd}} G_w / \sum_w G_w \leq 2 \sum_{w \text{ odd}} G_w / G_0$. For $|\phi| \leq \pi$ the odd term nearest ϕ gives $e^{-2\pi(\pi-|\phi|)/t}$, the opposite one at most the same, and the terms $|w| \geq 3$ are smaller by $e^{-12\pi^2/t}$; for $t \leq 1$ the total is below $5e^{-2\pi(\pi-|\phi|)/t}$. $\square$

Equation (7) displays the parallel move for one cube with N1–N3 absent (the group is abelian and flat): the mid-face weight is the heat kernel at the *softened* heat time $t_m + \frac{1}{4} \sum t_j$, the isolated-cube counterpart of the noise $v(k)$ in (5), evaluated at the interpolated flux. The only departure is the odd-charge factor, which differs from 1 only where a side-face flux approaches the cut locus $|\phi| = \pi$, where its bridge midpoint becomes bimodal. That is N4, exactly, with the rate $e^{-2\pi(\pi-|\phi|)/t}$. For a whole mid-plane, Poisson summation on each edge turns Ψ into a sum over integer windings of Gaussian terms, a Coulomb-gas representation in which the same rate controls the winding activities; that computation is left open. For $SU(2)$ and $SU(3)$ the bridge expectations $E R(m_j)$ are matrices, and the isolated cube is where N2 and N3 first appear.

Hypothesis P(α) (one parallel step, $D = 3$). There exist $\alpha > 0$, $t_0 > 0$, $C < \infty$ and a local quadratic form $\mathcal{D}_0$ (the free defect (5), possibly with renormalized t_{23}), such that for $t = \lambda_3 a \leq t_0$, on the small-field set where every plaquette satisfies $|\log U_p| \leq t^{1/2-\delta}$,

$$\left| -\log \Psi(U) - \mathcal{D}_0(U) - c_{\text{vac}} N_{\text{cells}} - \delta_t S_{23}(U) \right| \leq C t^\alpha \sum_{\text{cells}} (1 + |\log U_p|^2/t),$$

with a coupling shift $|\delta_t| \leq Ct$, and with a large-field suppression of Ψ by at least $e^{-ct^{-2\delta}}$ per violating plaquette, relative to its small-field value.

Under P(α) the per-step error on smooth observables is $O((\lambda_3 a)^\alpha)$ and the coupling shifts sum to a finite renormalization, because $\sum_n \lambda_3 a_n = 2\lambda_3 a_0$: the superrenormalizable pattern of the refinement note’s table, now with the term to be bounded identified exactly. P(α) is the directional form of what Balaban’s three-dimensional ultraviolet stability controls in his block-averaging scheme (Balaban 1985, CMP 102, 255, metadata); whether his estimates transfer to the heat-kernel directional scheme is unverified. P(α) yields ultraviolet control and says nothing about the gap: the long-distance spectral estimate of the refinement note §7 remains a separate obligation.

5. Four dimensions

In $D = 4$ a halving of direction 1 inserts parallel copies in the three planes (23), (24), (34), and the mid-space is a three-dimensional lattice gauge theory with bridge-distributed edges. By the table in §2 the heat time returns to itself after an isotropic step, so the whole one-loop running $g_0^{-2}(2a) = g_0^{-2}(a) - 2b_0 \log 2 + O(g_0^2)$, with $b_0 = 11/(16\pi^2)$ for $SU(3)$ in the convention $g_{\text{lat}}^2 = \lambda_4$, must come out of the four Ψ factors of one isotropic step. The N3 fluctuation term is therefore order one in $D = 4$ (it carries the logarithm), while in $D = 3$ it is $O(\lambda_3 a)$. This is the exact location of marginality in the insertion language. The corresponding hypothesis is H1 of the conditional theorem, and Balaban’s four-dimensional work (Balaban 1987, metadata) is the established partial control.

6. Consequence for STATE

Item 1 of STATE is answered in its “exact obstruction” form. The one-step 1+2 gauge refinement factors exactly into series moves, which close, and a parallel insertion Ψ (Proposition 1). Ψ is computed in closed form for the free field (Proposition 2, bound (6)), and the interacting

estimate is stated as Hypothesis $P(\alpha)$ with its four terms; Proposition 3 settles the isolated cube for $U(1)$. The next concrete step on this line is to prove $P(\alpha)$ for **one isolated cube with $G = SU(2)$, then $SU(3)$** (a single mid-face, where independence of the four bridge variables reduces Ψ to $\sum_R d_R e^{-t_m C_2(R)/2} \text{tr}_R \prod_j E R(m_j^{\pm 1})$ with t_m the mid-face heat time and the sign fixed by orientation), and then for a mid-plane with small-field cluster expansion. For the paper, the series/parallel split is the organizing statement of the dimension comparison: Newton's time insertion is a series move closed by one scalar; two-dimensional gauge theory has only series moves; three and four dimensions add $\binom{D-1}{2}$ parallel insertions whose elimination is the renormalization problem.